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3.) 1.) Dik: $F(x) = 5 + 2x^2$

$$W = \int_0^4 F(x) dx$$

Metode persegi titik tengah

$$a = 0, b = 4, n = 4$$

$$\Delta x = \frac{b-a}{n} = \frac{4-0}{4} = \frac{4}{4} = 1$$

Titik tengah setiap sub-interval:

$$x_i^* = a + \left(i - \frac{1}{2}\right) \Delta x$$

$$\Delta x = 1 ; x_1^* = 0,5$$

$$x_2^* = 1,5$$

$$x_3^* = 2,5$$

$$x_4^* = 3,5$$

$$\bullet F(x) = 5 + 2x^2$$

$$x_1^* = 0.5$$

$$F(0.5) = 5 + (0.5)^2 = 5 + 0.5 = 5.5$$

$$\bullet x_2^* = 1.5$$

$$F(1.5) = 5 + 2(1.5)^2 = 9.5$$

$$\bullet x_3^* = 2.5$$

$$F(2.5) = 5 + 2(2.5)^2 = 5 + 12.5 = 17.5$$

$$\bullet x_4^* = 3.5$$

$$F(3.5) = 5 + 2(3.5)^2 = 5 + 24.5 = 29.5$$

Pendekatan Integral Metode Persegi Titik Tengah:

$$W \approx \Delta x [F(x_1^*) + F(x_2^*) + F(x_3^*) + F(x_4^*)]$$

$$W \approx 1 [5.5 + 9.5 + 17.5 + 29.5]$$

$$W \approx 62 \text{ joule}$$

2.) Dik : $C(T) = 20 + 0,05 T^2$

$$Q = \int_0^{100} C(T) dT$$

Metode Trapezium

$$a = 0, b = 100, n = 4$$

$$\Delta T = \frac{b-a}{n} = \frac{100-0}{4} = \frac{100}{4} = 25$$

Titik Partisi : $T_i = a + i\Delta T$

$$T_0 = 0$$

$$C(T) = 20 + 0,05 T^2$$

$$T_3 = 75$$

$$T_1 = 25$$

$$T_0 = 0$$

$$C(75) = 20 + 0,05(75)^2$$

$$T_2 = 50$$

$$C(0) = 20 + 0,05(0)^2$$

$$= 301,25$$

$$T_3 = 75$$

$$= 20$$

$$T_4 = 100$$

$$T_1 = 25$$

$$T_4 = 100$$

$$C(25) = 20 + 0,05(25)^2$$

$$C(100) = 20 + 0,05(100)^2$$

$$= 51,25$$

$$= 520$$

$$T_2 = 50$$

$$C(50) = 20 + 0,05(50)^2$$

$$= 145$$

Pendekatan Integral Metode Trapezium :

$$Q \approx \frac{\Delta T}{2} [C(T_0) + 2C(T_1) + 2C(T_2) + 2C(T_3) + C(T_4)]$$

$$Q = \frac{25}{2} [20 + 2(51,25) + 2(145) + 2(301,25) + 520]$$

$$Q = 12,5 [20 + 102,5 + 290 + 602,5 + 520]$$

$$Q = 12,5 [1,535]$$

$$Q = 19187,5 \text{ J}$$

3.) Dik : $I_t = 2 + 3t + t^2$
 $q = \int_0^6 I(t) dt$

Metode Simpson $\frac{1}{3}$
 $a = 0, b = 6, n = 6$

$$\Delta t = \frac{b-a}{n} = \frac{6-0}{6} = 1$$

Titik Partisi : $t_i = a + i \Delta t$

$$t_0 = 0$$

$$t_1 = 1 \quad t_5 = 5$$

$$t_2 = 2 \quad t_6 = 6$$

$$t_3 = 3$$

$$t_4 = 4$$

$$t_0 = 0$$

$$\bullet I(t_0) = 2 + 3(0) + (0)^2 = 2$$

$$\bullet t_1 = 1$$

$$I(1) = 2 + 3(1) + (1)^2 = 6$$

$$\bullet t_2 = 2$$

$$I(2) = 2 + 3(2) + (2)^2 = 12$$

$$\bullet t_3 = 3$$

$$I(3) = 2 + 3(3) + (3)^2 = 20$$

$$\bullet t_4 = 4$$

$$I(4) = 2 + 3(4) + (4)^2 = 30$$

$$\bullet t_5 = 5$$

$$I(5) = 2 + 3(5) + (5)^2 = 42$$

$$\bullet t_6 = 6$$

$$I(6) = 2 + 3(6) + (6)^2 = 56$$

pendekatan Integral Simpson $\frac{1}{3}$

$$q = \frac{\Delta t}{3} [I(t_0) + 4I(t_1) + 2I(t_2) + 4I(t_3) + 2I(t_4) + 4I(t_5) + I(t_6)]$$

$$q = \frac{1}{3} [2 + 4(6) + 2(12) + 4(20) + 2(30) + 4(42) + 56]$$

$$q = \frac{1}{3} [414] = \frac{414}{3} = 138 \text{ coulomb}$$